

Returns on Gold and Stock Market: A Bivariate Time Series Analysis

Pushkar Kale (1825), Mrunmayi Naik (1836)

INTRODUCTION

According to different researchers, the purpose of the investment is the way of saving in the future. In the Indian commodity market, gold has its exceptional importance due to its societal and cultural significance and is still considered as a safe investment when compared to equity and constantly rising because of its big demand in the country.

Gold prices are a good indicator of how healthy the Indian economy is. Investors gravitate towards gold when they are defending their investments from either a crisis or inflation and when the price of gold falls that generally means the market is strong and healthy because investors have gone for gold instead of other, more profitable investments such as stocks, bonds, real estate business and other investments.

Different researchers have proved that there is inverse relationship between gold prices and stock exchange. In the condition of the downturn economy, most of the investors invest into the gold.

This project aims to establish relation between the returns on gold and the returns in NSE, using Multivariate Time Series.

DATA

The data is downloaded from the website Investing.com . Monthly data of Returns on gold (%) and Returns in NSE (%) over the period Nov 2000 - Oct 2019 is considered.

Visualising the data and Preliminary tests

Both the series are plotted individually.

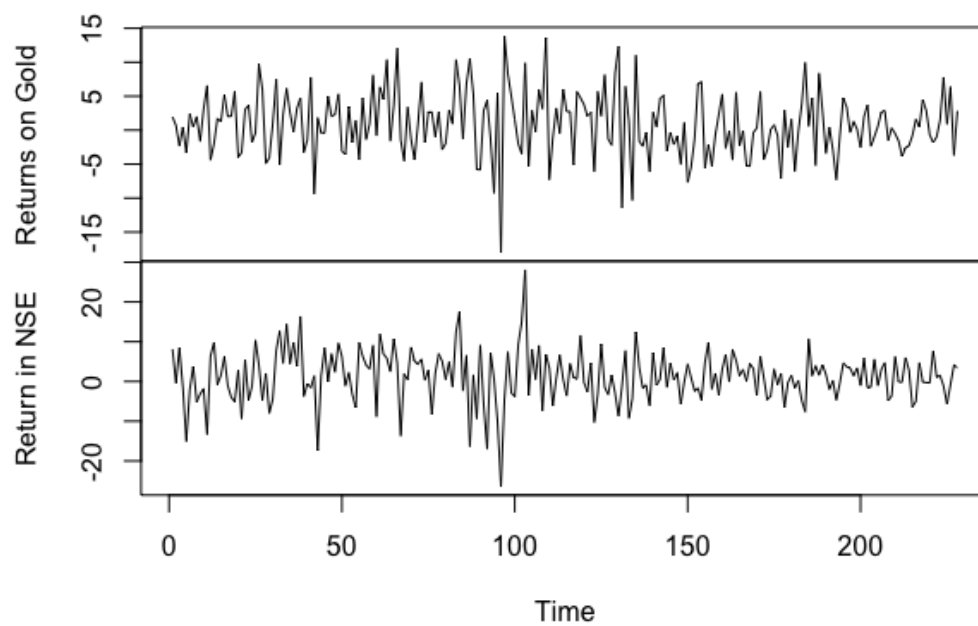


Figure 1

Both the series seem to have constant mean and constant variance with one or two outliers.

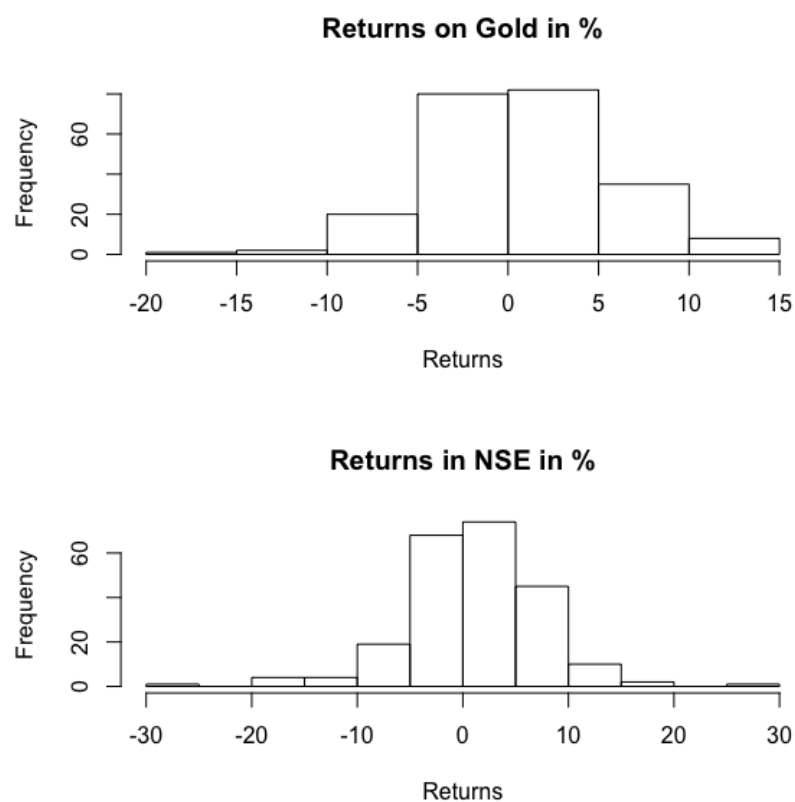


Figure 2

There is not enough evidence of non-normality for both the series. Hence, we can say that both the series are approximately normal. Following test confirms this –

```
One-sample Kolmogorov-Smirnov test
```

```
data: D_TS[, 2]  
D = 0.059406, p-value = 0.3969  
alternative hypothesis: two-sided
```

```
One-sample Kolmogorov-Smirnov test
```

```
data: D_TS[, 1]  
D = 0.038425, p-value = 0.8894  
alternative hypothesis: two-sided
```

Also, both the series are individually stationary (See below).

```
Augmented Dickey-Fuller Test
```

```
data: D_TS[, 2]  
Dickey-Fuller = -5.9438, Lag order = 6, p-value = 0.01  
alternative hypothesis: stationary
```

```
Augmented Dickey-Fuller Test
```

```
data: D_TS[, 1]  
Dickey-Fuller = -5.8205, Lag order = 6, p-value = 0.01  
alternative hypothesis: stationary
```

Hence, we would assume joint stationarity of the bivariate series for our analysis.

Also, there is not enough evidence of trend in both the series. Thus there is no need of differencing.

ACF and PACF of Series

Following are the ACF and PACF plots of individual series.

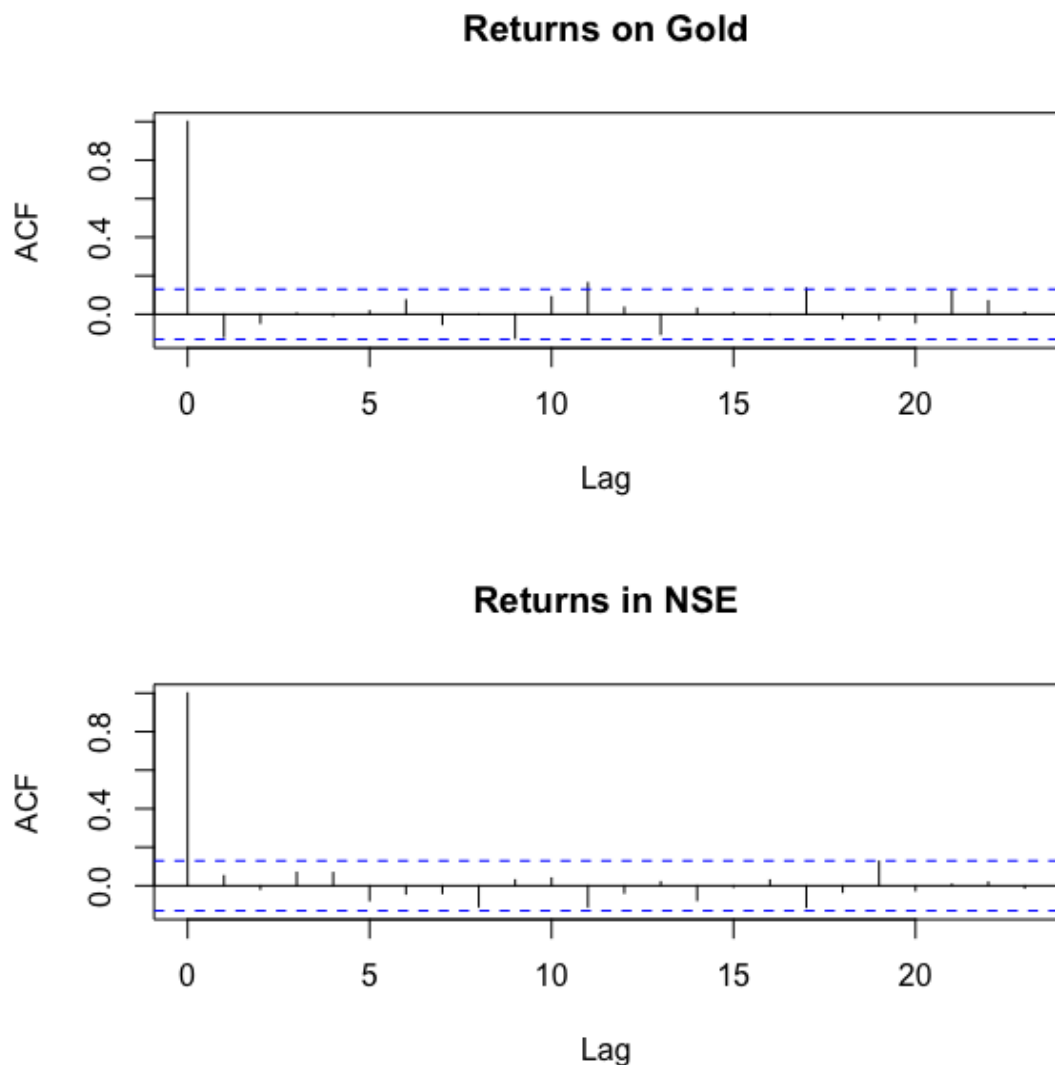


Figure 3

There is no significant spike at any of the lag. Hence, individually we can say that both the series are uncorrelated series. There is no clear pattern in PACF of both the series. For both the series all values of PACF are insignificant.

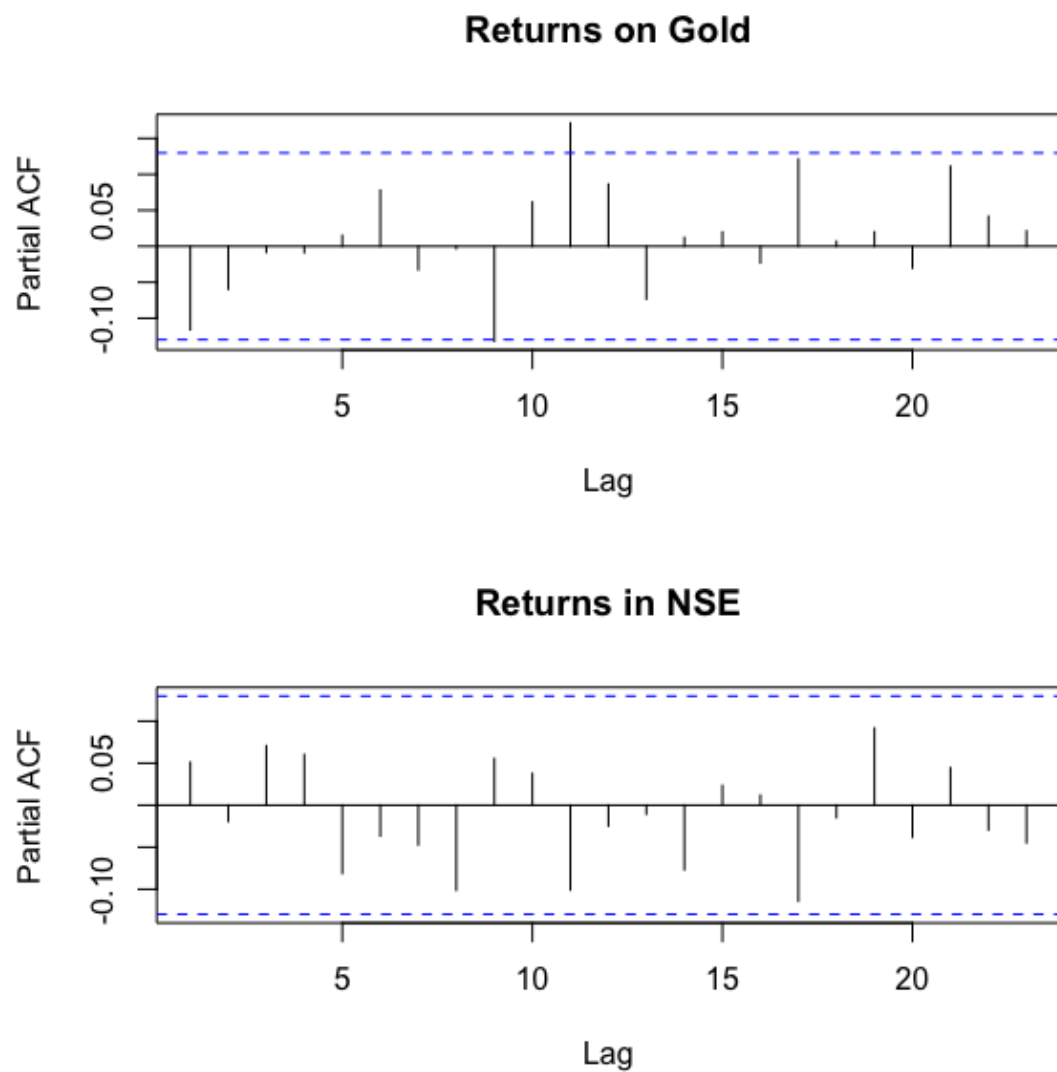


Figure 4

Now we will plot cross correlations of these series. There is no significant cross correlation is observed between the two series at any lag (See Fig 5).

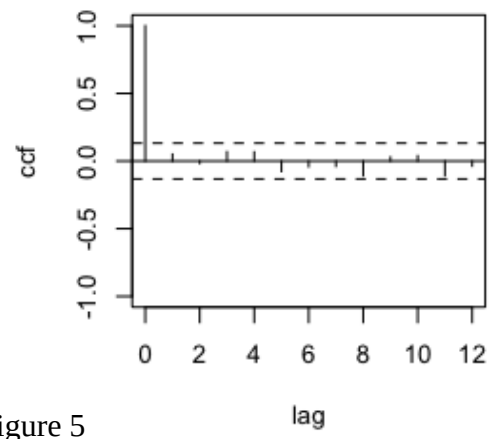
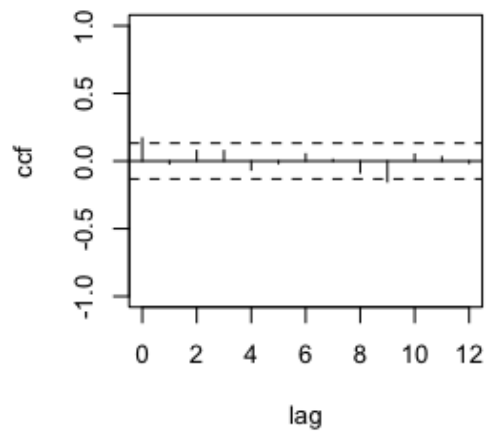
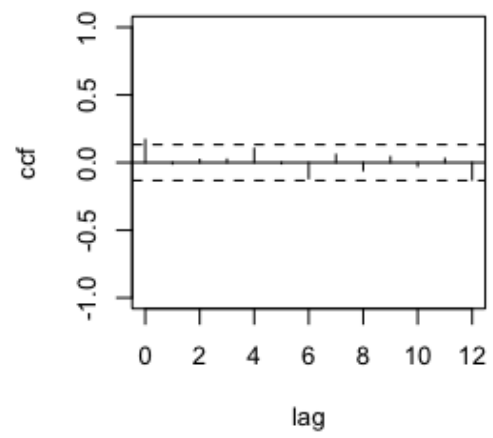
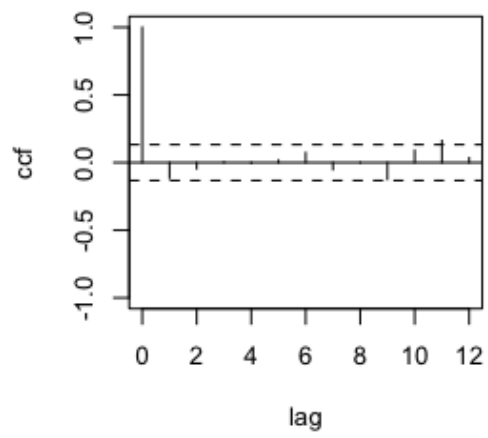


Figure 5

Fitting the Model

We have fitted following models. BICs of each model are calculated and the one with the minimum BIC value is chosen.

Model	VARMA(0,1)	VARMA(1,0)	VARMA(1,1)
BIC	6.979036	6.981345	7.063375

Table 1

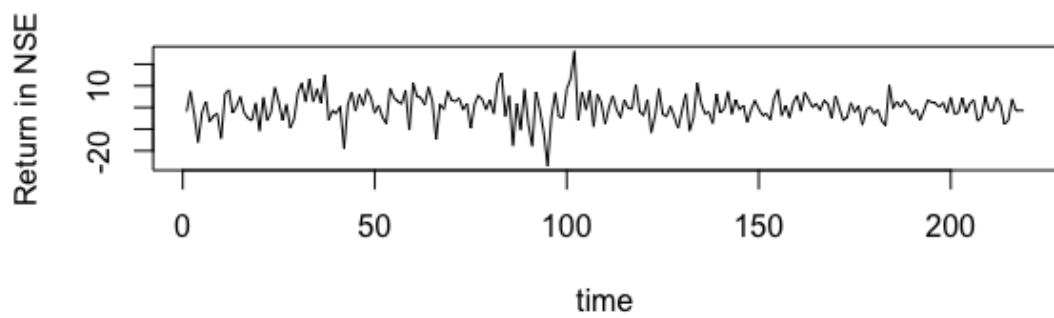
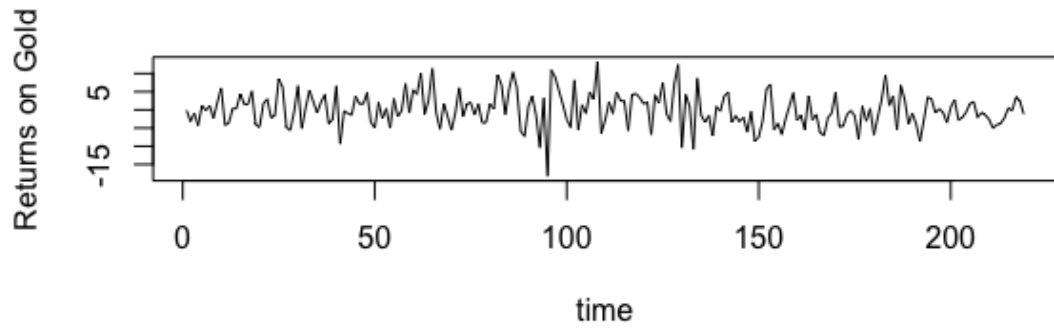
VARMA(0,1) is turned out to be the best model. Following are the details of the Model –

```
Estimates in matrix form:
Constant term:
Estimates:  0.8482636 1.197838
MA coefficient matrix
MA( 1 )-matrix
      [,1] [,2]
[1,] 0.1357 -0.0150
[2,] 0.0217 -0.0562

Residuals cov-matrix:
      [,1] [,2]
[1,] 22.328831 5.573451
[2,] 5.573451 42.906337
----
aic= 6.886483
bic= 6.979036
```

The Residual analysis is done for this model.

Residual Analysis



Residuals are plotted. In both the series, residuals don't show any specific pattern.

Ljung-Box Statistics:				
	m	Q(m)	df	p-value
[1,]	1.0000	0.0238	4.0000	1.00
[2,]	2.0000	3.1158	8.0000	0.93
[3,]	3.0000	4.6659	12.0000	0.97
[4,]	4.0000	10.3449	16.0000	0.85
[5,]	5.0000	11.7277	20.0000	0.93
[6,]	6.0000	17.4551	24.0000	0.83
[7,]	7.0000	19.1888	28.0000	0.89
[8,]	8.0000	23.8205	32.0000	0.85
[9,]	9.0000	31.7549	36.0000	0.67
[10,]	10.0000	35.5835	40.0000	0.67
[11,]	11.0000	48.2008	44.0000	0.31
[12,]	12.0000	54.2020	48.0000	0.25
[13,]	13.0000	62.6189	52.0000	0.15
[14,]	14.0000	66.7190	56.0000	0.15
[15,]	15.0000	70.4493	60.0000	0.17
[16,]	16.0000	72.2506	64.0000	0.22
[17,]	17.0000	80.1170	68.0000	0.15
[18,]	18.0000	82.1802	72.0000	0.19
[19,]	19.0000	87.2039	76.0000	0.18
[20,]	20.0000	88.7506	80.0000	0.24
[21,]	21.0000	94.9757	84.0000	0.19
[22,]	22.0000	97.4397	88.0000	0.23
[23,]	23.0000	98.5365	92.0000	0.30
[24,]	24.0000	101.5460	96.0000	0.33

p-values of Ljung-Box statistics

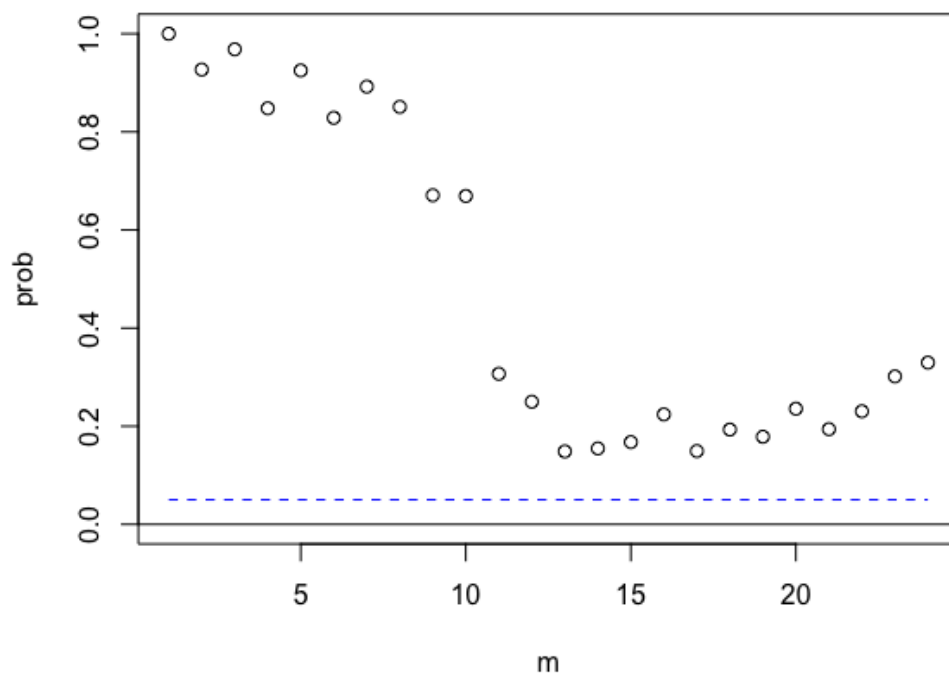


Figure 6

From Ljung-Box p-values for the bivariate residuals, we can see that all the p-values are greater than 0.05. Hence, residuals are independently distributed.

Individual Normalities of residuals are tested.

Anderson-Darling normality test

```
data: Model$residuals[, 2]  
A = 0.84478, p-value = 0.02923
```

Anderson-Darling normality test

```
data: Model$residuals[, 1]  
A = 0.37478, p-value = 0.412
```

At 1% level, normality of residuals is accepted for both the series.

From all the above conclusions, we can say that our model is good.

FORECASTING

	Returns on Gold	Return in NSE	Returns on Gold	Return in NSE
	-1.83	7.70	0.9838717	1.142617
fcst	-0.98	1.07	0.8482636	1.197838
fcst	1.46	1.49	0.8482636	1.197838
fcst	7.75	-1.12	0.8482636	1.197838
fcst	0.89	-5.69	0.8482636	1.197838
fcst	6.37	-0.85	0.8482636	1.197838
fcst	-3.69	4.09	0.8482636	1.197838
fcst	2.84	3.51	0.8482636	1.197838

Out of 228 observations, 220 are used for the fitting and remaining 8 are predicted. From above output, we can see that forecasts are not reasonable. Hence, we may need better method for the forecast.

CONCLUSIONS

We have found that VARMA(0,1) is the best model for the series Returns on Gold and NSE Returns. All the residual assumptions are satisfied. Forecasts have been made but they are not so reasonable. Improved forecasts using some advanced time series techniques would be desirable from the investment point of view.